

GCSE PAPER 2 PROBLEM SHEET A

Fifteen problems for Edexcel higher paper two (calculator).

1. (a) t is directly proportional to s . When $t = 3.5$, $s = 24.5$. Find the value of t when $s = 35$.

(b) c is inversely proportional to the square root of d . When $c = 1$, $d = 9$. Find the value of d when $c = 0.12$.

2. Consider a scatter plot of temperature ($^{\circ}\text{C}$) and ice cream sales, and suppose the scatter plot exhibits a positive correlation.

(a) Sketch an example of such scatter plot. You do not need to label your axes.

(b) Draw a line of best fit through your data.

(c) Explain how to use your line of best fit for future prediction.

3. Consider the graph $y = x^3 + 4x^2 - 6x - 3$.

(a) Complete the table below.

x	-3	-2	-1	0	1	2	3
y	24			-3			

(b) Use the table to sketch the shape of the graph for values of x between -3 and 3 . The sketch does not need to label any points of intersection with the axes.

4. Alice and Bob live in Hobbiton. In this world, the price of a chocolate bar has risen from £5 to £14 over the last five years, whereas the price of a sandwich has risen from £8 to £12.80.

Bob claims the percentage increase of the price of chocolate is three times the percentage increase of the price of sandwiches. Is he correct? Show your working.

5. A triangular prism has a cross-sectional area of 15cm^2 and a length of 12cm . It is made of chocolate with a density of 0.8g/cm^3 .

A cuboid has a width of 4cm , height of 5cm , and length of $l\text{cm}$. It is made of marshmallow with density 0.72g/cm^3 .

Both objects have the same mass. Find l .

6. Consider the first five terms of a quadratic sequence,

$$4, \quad 11, \quad 22, \quad 37, \quad 56.$$

(a) Find an expression for the n th term of this sequence.

(b) What is the 100th term of this sequence?

7. Simplify

$$\frac{2x^2 - 5x - 3}{x^2 - 9} \div \frac{4x^2 - 1}{2x^2 + 5x - 3}.$$

8. There are only red counters and blue counters in a bag. The probability a counter is red is 0.4.

Two counters are taken at random from the bag without replacement. The probability that both counters are red is $\frac{13}{85}$.

Calculate the number of blue counters originally in the bag.

9. (a) Toby invested £4500 into a savings account for 3 years. He was paid compound interest at a rate of 3.95% per annum.

Calculate the total amount of interest Toby earned after 3 years, giving your answer to the nearest penny.

10. A memory stick has a capacity of 64 gigabytes, correct to the nearest gigabyte. A video file has a size of 4.2 gigabytes, correct to the nearest 0.1 of a gigabyte.

Calculate the upper bound for the number of these video files that can be completely stored on the memory stick. Show your working.

11. A and B are two mathematically similar solid shapes. Shape A has a surface area of 54cm^2 and a volume of 324cm^3 . Shape B has a surface area of 96cm^2 .

Calculate the volume of shape B .

12. (a) Complete the table of values for $y = 2x^2 - 4x - 1$.

x	-1	0	1	2	3	4
y	5		-3		5	

(b) On a grid, draw the graph of $y = 2x^2 - 4x - 1$ for values of x from -1 to 4.

(c) Use your graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 2$.

13. Prove algebraically that $0.2\dot{4}\dot{5}$ can be written as the fraction $27/110$.

14. The table shows information about the speeds of 60 cars traveling past a speed camera.

Speed (v mph)	Frequency
$30 < v \leq 40$	12
$40 < v \leq 50$	22
$50 < v \leq 60$	18
$60 < v \leq 80$	8

- (a) Calculate an estimate for the mean speed of the cars.
- (b) Draw a frequency polygon for the information in the table.

15. Three consecutive terms in a geometric progression are

$$x - 14, \quad 18, \quad 2x + 14.$$

Find the two possible values of x for this sequence, also stating the two corresponding common ratios r .